Scalability & Future Plan

Date	27 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	1 Mark

Scalability & Future Plan

This document explains how the **Rising Waters** project can be expanded to support more users, larger datasets, and advanced features in the future. It also outlines the planned improvements that will enhance the system's performance, reliability, and usability.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Limited historical weather dataset	Prediction accuracy may decrease for unseen weather conditions	High
2	Supports prediction for a limited geographical area	Cannot provide reliable predictions for all regions	High
3	Single-server Flask deployment	Performance may reduce with many simultaneous users	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users	Deploy on cloud infrastructure with load balancing to support multiple concurrent users
2	Data Storage	Local CSV dataset	Integrate a cloud-based database such as MySQL or PostgreSQL to manage large datasets
3	Performance	Predictions generated using a locally hosted model	Optimize the model and use GPU/cloud resources for faster processing
4	Security	Basic input validation	Implement user authentication, HTTPS, secure APIs, and encrypted data storage

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Increase the dataset using recent weather and flood records	2 Months	Improves prediction accuracy and reliability
Phase 2	Integrate real-time weather data from APIs	4 Months	Provides live flood risk prediction based on current conditions
Phase 3	Develop an Android mobile application	6 Months	Makes the system more accessible to users and emergency responders
Phase 4	Implement GIS maps and SMS/Email alert notifications	8 Months	Enables location-based flood monitoring and early warning alerts